# Who Decides? System One Models in the Decision Seat of Clinical Agents

**Yubo Li** **[co-authors]**
Carnegie Mellon University
yubol@andrew.cmu.edu

## Abstract

Clinical agents read records, order tests and write notes, and every step they take is a decision generated by a large language model (LLM) after thousands of tokens of reasoning. System One decision models decide in a single forward pass: they read a state and a set of options and return a probability for each. We put six such models in the decision seat of a real clinical agent loop. On PhysicianBench, 100 consultation tasks on an electronic health record of real de-identified patients, the role matters more than the model. Choosing among five turns proposed by the LLM, a System One model is safe: Clef-27B passes 80.2% of checkpoints against 81.3%, with no significant change in task success. Choosing the tool itself, it breaks the agent (7.0% of tasks pass). Neither role improves on the LLM, because deciding is not where the LLM's errors are: 79% of its step-level "omissions" of a required order are placed later anyway, and only 11% of required orders are ever proposed but not executed, the most any selector can recover. A decider that recovers part of this loses as much by acting before the LLM has proposed the rest; an early gain on orders failed a pre-registered replication. What the System One models bring instead is a shared *commitment bias*: all six prefer turns that order (1.3–1.6× the LLM's rate) or finish (1.3–1.4×) over turns that gather information. A pre-registered blind test splits it in two: about two thirds of the gain on orders needs no patient state, a prior over the options, while the excess of premature finishing appears only when the model reads the state. We map where System One deciders are reliable (their confidence is informative; they help where the LLM's samples disagree; finishing is their blind spot), show that step-level answer keys measure decision style rather than outcome, and derive a division of labour in which the System One model decides only at order steps.

## 1 Introduction

A physician's agent takes dozens of steps per task: look up the problem list, pull the labs, read last month's note, order the imaging, write the assessment. Each step is a choice among a small set of concrete actions, and in today's systems an LLM makes every one of them by generating text (Liu et al., 2026; Yao et al., 2023). Figure 1 follows one such task.

*System One* decision models offer a different way to choose (Almeida, 2026; Cloudflare, 2026). They take a state and a typed question with named options and return a probability for every option in one forward pass, with no generated text and nothing to parse. They are fast, cheap and suited to choosing among options, the operation an agent performs at every step. Whether they can make an agent's decisions is open. So far they have been studied on single-shot tasks and as judges (Li et al., 2026b), and in general-purpose agent loops their actions often go wrong even when they read the state correctly (Li et al., 2026a). No study has placed one in the decision seat of a clinical agent.

We do so under a strict comparison. Clef-27B is post-trained from Qwen3.8-27B, which serves as our LLM, so the two decision makers share a backbone and differ in how they read out a decision. We run PhysicianBench in its own harness, with its own tools and graders, and change only the component that picks the next turn (Figure 2). Five further open System One models take the same seat on a fixed exam of logged decisions, and one of them online. We organise the study around three questions: how System One models perform in the loop, where their limits lie, and how they should share the work with the reasoning LLM. The short answer: they are safe as selectors and harmful as steerers, they cannot do much better than the LLM because the LLM's errors are mostly not in the choice of the next turn, and what they bring instead

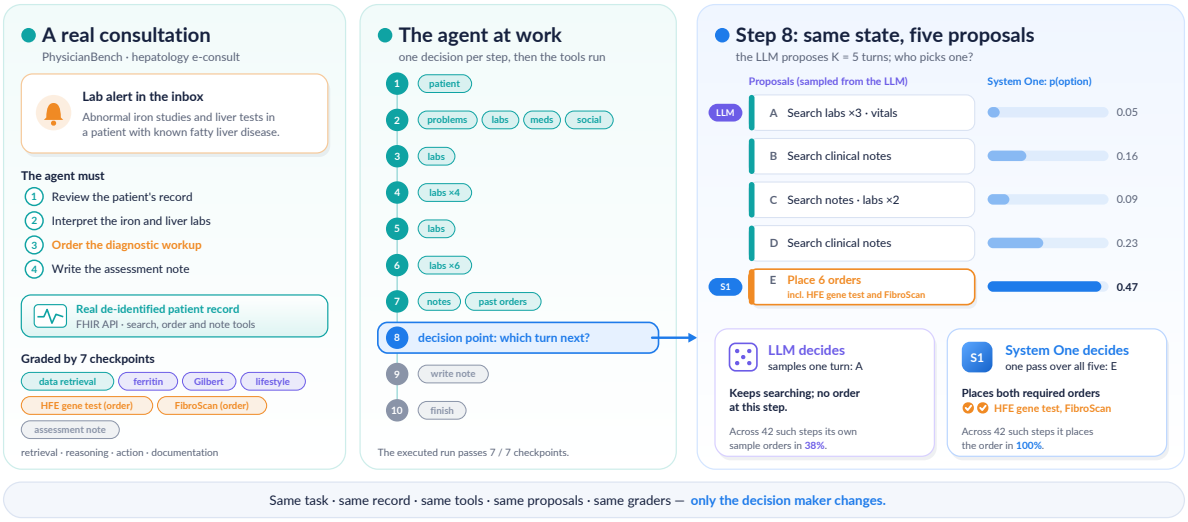



Figure 1: **One real consultation, one decision, two deciders.** A PhysicianBench task (instruction paraphrased): a hepatology e-consult on abnormal iron and liver tests, in which the agent must review the record, interpret the labs, order the workup and write a note, graded by seven checkpoints. At step 8 the LLM has proposed five next turns. Its own sample (A) keeps searching labs. The System One model reads the same state and all five proposals in one forward pass and puts the most probability on E, which places both orders the graders require. Across all 42 order steps where a proposal places a required order, the LLM's own sample orders in 38% and Clef in 100%; most of the LLM's misses are placed at a later step (§4.2). The figure shows tool types, probabilities and outcomes only, never patient data.

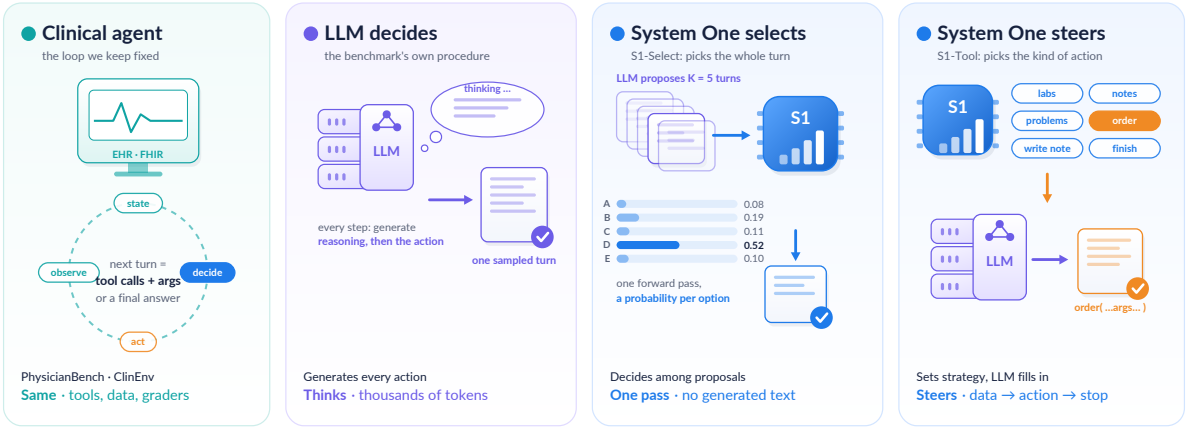



Figure 2: **Ways to make the next decision in the same clinical agent loop.** The loop, tools, records and graders stay fixed. LLM: the model samples its next turn after reasoning. S1-SELECT: the LLM proposes five complete turns and a System One model picks one in a single forward pass. S1-TOOL: the System One model picks the kind of action from the full tool menu and the LLM writes the call. A gated variant of S1-SELECT (§2) consults the System One model only at order steps.

is a shared bias toward committing, part prior and part misreading of the state, which tells us where to use them.

**Findings.**

- **The role matters more than the model** (§4.1, §4.4). Choosing among proposals, Clef matches the LLM on checkpoints (−1.1 [−4.7, +2.7] points) and neither System One model changes task success significantly; choosing the tool, task success falls to 7.0%.

- **Deciding is not the bottleneck** (§4.2). 79% of the LLM's step-level omissions of a required order are placed later in the same run. Of all required orders, 11% are proposed but never executed, the most any selector can recover, and 19% are never proposed. pplx-decider-v1.2 re-

covers part of the 11% and loses as much on generation; its early +13 points on required orders failed a pre-registered replication on 24 held-out tasks ($-18$ points, 95% CI $-36$ to $-2$).

- **They carry a commitment bias** (§4.3). All six models prefer turns that order or finish over turns that gather information. A rule with no model that always takes an ordering turn when one is offered reaches most of their advantage on the order key; their apparent avoidance of redundant steps is mostly finishing instead of rewriting the note. Asked blind, without the patient state, pplx-decider-v1.2 keeps about two thirds of its gain on orders but finishes early less often than chance instead of more: the order preference is largely a prior, the excess of premature finishing comes from reading the state.
- **Steering alone, the bias breaks the agent** (§4.4). Choosing the tool, Clef cuts retrieval before the first action from 30.1 to 9.9 calls and writes the note 9.2 times per task.
- **Offline exams measure style** (§4.2). Every decider beats the LLM's own sample on a fixed exam of logged decisions (0.17–0.27 against 0.48 key errors), but the step-level key rewards ordering early, and the exam holds too few consequential decisions to rank deciders on outcome.
- **Where a System One decider is reliable** (§4.5). Its confidence is informative; it helps where the LLM's five samples disagree and adds nothing where they agree; it is confidently wrong about when to finish.
- **A narrower division of labour** (§4.7). Consulting the System One model only where a proposal places an order keeps the decider where its prior helps and away from stopping, and overrules the LLM on 11% of steps instead of 73%. Its pre-registered online test is under way.

## 2 Putting a System One model in the decision seat

An agent loop alternates between a state $s_t$, the task plus every action and result so far, and the next turn $a_t$: a set of tool calls with arguments, or a final answer. The benchmark samples $a_t \sim \pi(\cdot \mid s_t)$ from the LLM. We replace only this step.

**LLM (reference).** One sample from $\pi$ with thinking on and the model's default sampling, exactly as the benchmark prescribes. A logging variant samples $K{=}5$ turns and executes the first, which has the same distribution as LLM and gives the run-to-run noise floor.

**S1-SELECT: the System One model chooses the turn.** The LLM samples $K{=}5$ turns $a_t^{(1)}, \ldots, a_t^{(5)}$ i.i.d. from $\pi$ in one request. Exact duplicates are merged. The System One model reads the state and the distinct turns, rendered as tool calls with full arguments, and returns a probability for each; the most probable turn is executed. It never sees the LLM's reasoning, so it judges actions, not rationales. A uniform pick among $K$ i.i.d. samples has the same distribution as LLM, so any difference between the two comes from *which* sample is chosen.

**S1-TOOL: the System One model chooses the kind of action.** The System One model reads the state and the full tool menu (each tool's name and description, plus FINISH) and picks one. The LLM then writes the turn with its calls constrained to the chosen tool. The System One model now sets the agent's strategy and timing, and the LLM supplies the content. LLM-TOOL is the control: the same one-tool-per-turn restriction with the LLM choosing (it keeps only the calls of its own turn's first tool).

**Gated selection: System One plus the reasoning LLM.** The System One model is consulted only at some steps and only overrules the LLM's own first sample there. In the variant we test online it is consulted only where at least one proposal places a service or medication order; elsewhere, including every choice between finishing, writing the note and gathering more data, the LLM's first sample is executed. A margin variant overrules only when the System One model prefers its pick to the LLM's sample by a log-odds margin.

**Fair inputs.** Options are named by letters shuffled with a seed per task, run and step. The System One model reads a deterministic 32k-token view of the state (oldest tool results shortened first); the LLM reads up to 128k tokens. If the System One model fails, the first sample is executed; this happened 0 times in the S1-SELECT run.

## 3 Setup

**PhysicianBench** (Liu et al., 2026) adapts 100 real outpatient consultations from 21 specialties into long-horizon tasks on a FHIR server with the

records of 108 de-identified patients (47k conditions, 114k observations, 35k procedures, 13k medication requests, 2.3k clinical notes). Each task has 5–9 checkpoints (670 in total) of four types: data retrieval (DR), clinical reasoning (CR), action execution (AE: orders, prescriptions, referrals placed in the record and verified by code) and documentation (Doc). A task passes only if all of its checkpoints pass. We use the released harness: a fresh server and database per task, the MiniAgent loop with parallel tool calls and at most 100 steps, and the task's pytest graders. For analysis we split AE into the 92 code-graded required-order checkpoints in 54 tasks (the graders' own name and code patterns), and the rest. Only one checkpoint in 100 tasks penalises an order ("must not be ordered"), so the benchmark barely penalises over-ordering.

**Models.** The LLM is Qwen3.8-27B (Qwen Team, 2026) in bf16 with 128k context and thinking on, served by vLLM (Kwon et al., 2023). The main System One model is Clef-27B (Cloudflare, 2026) with its native joint head. The other five are pplx-decider-v1.2, d3, linne-oak-1.0, amos-1-large and decider-31b, each served by its authors' code at a pinned revision (Appendix A). The judge is the same Qwen model with thinking off, shared by all arms. Everything runs on one node with four H100 GPUs; no record leaves it (§6).

**The fixed exam.** To compare deciders on identical inputs, the LLM runs every task once and logs, at each step, its five proposals and the full state. Each step with at least two distinct proposals is a question (1466 questions from 100 tasks), and every decider answers byte-identical requests: the same state, the same options, the same option letters. Three keys apply where they separate the options: *order* (the option places a required order not yet placed), *finish* (finishing while a required order is still open is premature) and *redundancy* (the option repeats executed calls, or rewrites the note with no new tool result).

**Statistics.** Arms are compared on the tasks both completed. We report differences with 95% bootstrap intervals over tasks (10,000 resamples for single runs, 5,000 for pooled comparisons) and exact McNemar tests for task success. Where several runs of the same policy exist (LLM run 1, LLM run 2 on 15 tasks, and the logging variant; 2.15 runs per task on average) we pool them per task before pairing.

# 4 Results

## 4.1 Performance: choosing among proposals

Table 1 and Figure 3 give the headline for Clef. S1-SELECT passes 80.2% of checkpoints against 81.3% for the LLM ($\Delta$ −4.0 [−13.0, +5.0] points in task success, McNemar $p = 0.52$). The noise floor is wide: two runs of the same LLM policy differ on 28 of 100 tasks ($\Delta$ +0.0 [−10.0, +10.0] points), so single-run differences in task success of a few points carry no signal. Pooling the LLM's runs sharpens the checkpoint-level comparison (Table 2).

Neither System One model moves the checkpoints. An early gain of pplx-decider-v1.2 on required orders failed a pre-registered replication (§4.2). Both shorten the episode: they act after less retrieval, write fewer notes and take fewer steps. Documentation checkpoints dip by 2–4 points, within noise. Nothing in our data establishes a gain in task success.

## 4.2 Deciding is not the bottleneck

Why does a better pick not show up in the outcome? On the fixed exam every decider makes far fewer key errors than the LLM's own sample (0.17–0.27 against 0.48; §4.3), yet online the checkpoints do not move. Following each required order along the LLM's own trajectories resolves the gap (Table 3). Of the 94 steps at which the LLM's sample skipped an available required order, 79% are followed by that order later in the same run: the step-level key mostly rewards ordering early, not ordering at all. Of all 118 required orders in 54 tasks, the LLM places 69%, 11% are proposed at some step but never executed, and 19% are never proposed by any sample. Selection can at most recover the 11% (on these trajectories, 4 more of 54 tasks would pass); the 19% are bounded by generation. The decision seat holds little of the LLM's error.

pplx-decider-v1.2 shows what a selector does with that margin. It recovers part of it (proposed but never executed: 4% against 11%, −7 points [−13, 0]) and gives about as much back on generation: it acts after less retrieval, and 4 points more of the required orders are never proposed ([−5, 13]). It places about as many as the LLM (+3 [−8, 12]). Its apparent gain on its first 76 tasks (+13 points on required-order checkpoints) failed a pre-registered replication on the 24 tasks it had not run (−18 [−36, −2], against an LLM that already passes 84% of them).

| | Task pass@1 | | All checkpoints | | Checkpoints by type (%) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Decision maker | % | Δ [95% CI] | % | Δ [95% CI] | DR | CR | AE | Doc | Calls /task | Tokens /task |
| LLM | **41.0** | *ref.* | **81.3** | *ref.* | 83.2 | **84.1** | 75.0 | **70.7** | 36.3 | 29k |
| S1-SELECT | 37.0 | −4.0 [−13.0, +5.0] | 80.2 | −1.1 [−4.7, +2.6] | **85.0** | 82.8 | **80.1** | 65.7 | 31.2 | 119k |
| S1-TOOL | 7.0 | −34.0 [−45.0, −23.0] | 44.3 | −37.0 [−44.5, −29.3] | 44.7 | 47.5 | 52.3 | 26.3 | 25.7 | 18k |

Table 1: **PhysicianBench, run 1, 100 tasks completed by all three decision makers** (Clef as the System One model). Δ: difference from LLM in points with a 95% task bootstrap interval. Checkpoint types: data retrieval (DR), clinical reasoning (CR), action execution (AE), documentation (Doc). Calls: tool calls per task. Tokens: generated tokens per task, including all five proposals for S1-SELECT. Bold: best per column.

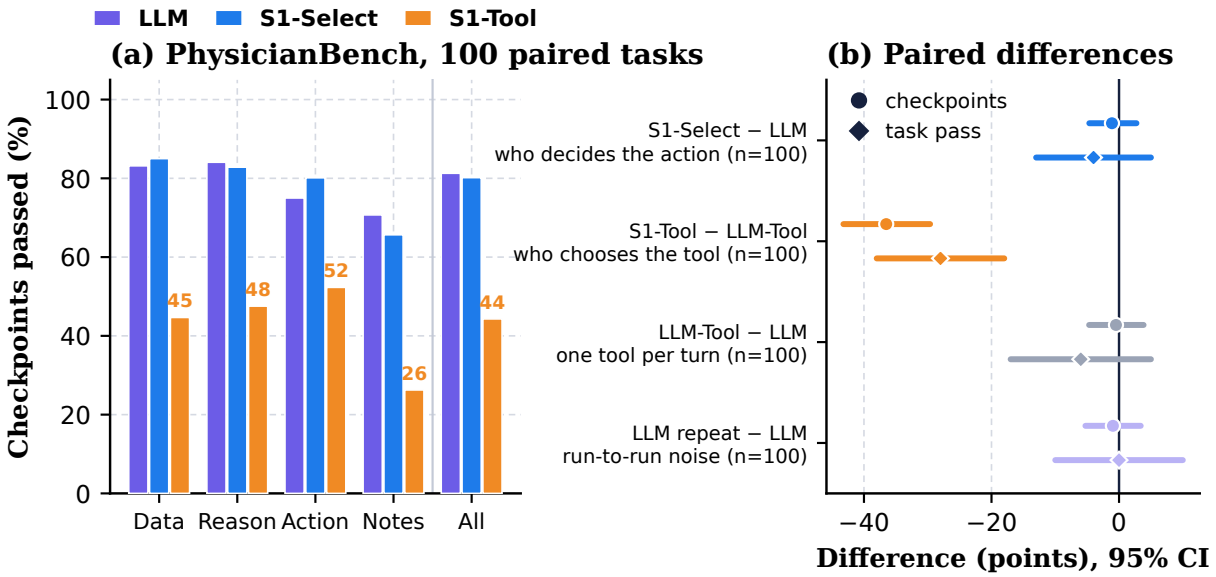



Figure 3: **A System One model keeps the LLM's quality when it chooses among proposals, and falls behind when it steers.** (a) Checkpoint success by type on the paired tasks. (b) Paired differences with 95% intervals, each on the tasks both arms completed.

| | LLM (pooled) | Clef (100 tasks) | pplx-decider (100 tasks) |
|---|---|---|---|
| Task pass | 0.41 | −0.04 [−0.12, 0.03] | +0.04 [−0.06, 0.14] |
| Required-order cp. | 0.76 | +0.01 [−0.06, 0.07] | +0.04 [−0.05, 0.14] |
| Documentation cp. | 0.70 | −0.04 [−0.11, 0.03] | +0.03 [−0.05, 0.12] |
| Required orders placed | 0.68 | −0.02 [−0.10, 0.05] | +0.05 [−0.05, 0.15] |
| Notes written | 1.5 | **−0.38** [−0.60, −0.09] | **−0.52** [−0.68, −0.37] |
| Retrieval before 1st action | 30 | **−5.5** [−8.0, −3.1] | **−3.2** [−5.6, −1.0] |
| Steps | 36 | **−5.1** [−8.1, −1.9] | **−4.0** [−6.7, −1.5] |

Table 2: **S1-SELECT against the pooled LLM runs**, paired by task (differences with 95% task bootstrap intervals; LLM column: the level on the 100 paired tasks). cp.: checkpoint pass rate. Bold: interval excludes 0.

| Required orders (%) | LLM | Clef | pplx |
|---|---|---|---|
| Placed | 69 | 64 | 72 |
| Proposed, never executed | 11 | 8 | 4 |
| Never proposed | 19 | 28 | 24 |
| Step omissions placed later | 79 | 13 | 25 |

Table 3: **Where required orders are lost,** along each arm's own trajectories on the 54 tasks with required orders (118 orders; LLM: the logging run). The middle row is the most a selector could recover.

The exam cannot see any of this. Restricted to steps where the counterfactual is known (the LLM's sample takes no open required order and a proposal places one the run never places), it holds only 11 questions in 9 tasks, too few to rank deciders (take rates 0.64–0.91, intervals ±0.2–0.3). Step-level answer keys inside an agent's own trajectory measure decision *style*; only whole runs rank deciders on outcome.

### 4.3 Mechanism: a commitment bias

Why do the System One models shorten the episode? The fixed exam answers on identical inputs (Tables 8 and 4). All six models agree with the LLM's own sample about as often as a random pick would (21–27% for five of them, 36% for linne-oak-1.0; random 22%), and with each other more often. They make far fewer key errors than the LLM's sample (0.17–0.27 against 0.48; every bootstrap in-

| | Error (3 keys) | Omits required | Over-orders | Finishes early | Redun-dant |
|---|---|---|---|---|---|
| LLM (own sample) | 0.48 | 55 | 35 | 80 | 28 |
| Always order (no model) | 0.22 | 0 | 100 | 77 | 25 |
| Clef-27B | 0.19 | 2 | 66 | 90 | 8 |
| pplx-decider-v1.2 | **0.17** | 3 | 71 | 90 | 4 |
| d3 | 0.18 | 4 | 67 | 90 | 4 |
| linne-oak-1.0 | 0.21 | 5 | 53 | 97 | 9 |
| amos-1-large | 0.27 | 18 | 53 | 90 | 11 |
| decider-31b | 0.21 | 7 | 52 | 93 | 8 |

Table 4: **Error profile on the fixed exam** (1,460 questions answered by all six, 100 tasks; 200 keyed: 97 order, 30 finish, 85 redundancy). Error: bad picks over all (question, key) pairs. Other columns in %: omits a required order on offer; orders when none is required among mixed options; finishes while a required order is open; picks a redundant option. Always order: take the most-sampled ordering option whenever one is offered.

terval excludes 0), but the errors move rather than vanish: the LLM omits an available required order at 55% of such steps and the deciders at 2–18%, while the deciders order when no required order is on offer at 52–71% of steps against the LLM's 35%, and finish while a required order is open at 90–97% of such steps against 80%.

Two observations show that much of this is a preference over kinds of action rather than a reading of the patient. First, a rule with no model, *always take an ordering turn when one is offered*, reaches an error of 0.22 and leaves no required order behind; on the order key it is better than every decider. Second, the deciders' low redundancy is not evidence that they track what has been done: in 17–18 of the 19–22 questions where a decider avoids the LLM's redundant option, it does so by choosing FINISH over a rewrite of the note. Table 5 puts the preference in one number per kind: every model picks ordering turns 1.3–1.6× and finishing turns 1.3–1.4× as often as the LLM's own sample does, and retrieval turns 0.89–0.93×. The same profile shows online (Table 2: less retrieval before acting, fewer notes, shorter runs). We call it a *commitment bias*: System One deciders prefer turns that commit (order, write, finish) over turns that gather information.

The bias is useful exactly where the LLM undercommits, and it costs elsewhere. pplx-decider-v1.2 online takes more of the required orders that a proposal offers (offered but never executed: 4% of required orders, against 11% for the LLM; difference $-7$ points $[-13, 0]$), but 4 points more are never proposed at all ($[-5, 13]$), so it places about as many as the LLM (+3 $[-8, 12]$). Clef does not even gain the first part: on its own trajectories it

| | Order | Finish | Note | Retrieve |
|---|---|---|---|---|
| Clef-27B | 1.57 | 1.34 | 1.06 | 0.89 |
| pplx-decider-v1.2 | 1.49 | 1.35 | 1.09 | 0.90 |
| d3 | 1.48 | 1.37 | 1.09 | 0.90 |
| linne-oak-1.0 | 1.37 | 1.43 | 1.18 | 0.89 |
| amos-1-large | 1.29 | 1.30 | 1.06 | 0.93 |
| decider-31b | 1.36 | 1.41 | 1.18 | 0.89 |

Table 5: **The commitment bias.** Picks of each kind of turn on the exam, relative to the LLM's own sample on the same questions ($>1$: chosen more often than the LLM chooses it).

places 6 points fewer of the required orders than the LLM ($[-15, 3]$), and 8 points more of them are never proposed at all ($[-1, 18]$), plausibly because it moves to action after less retrieval and the LLM then never thinks of them.

**A prior and a misreading (blind test, preregistered).** Does the bias read the patient at all? We asked pplx-decider-v1.2 the whole exam again with the options but no state (same option letters; test P4, fixed before its data existed). The preregistered style hypothesis, that blind and informed picks agree on at least 60% of questions and the order and finish errors move by at most 0.05, is rejected: they agree on 41% (random 22%), and blind picks match the kind of the informed pick barely more often than chance (91% against 88%; other deciders with the state: 95–98%). The model reads the state. But the keys move in opposite directions (Table 6). On orders, the blind model is already far better than chance: about two thirds of the reduction from a random pick needs no patient state, so the order preference is largely a prior over what the options look like. On redundancy, the state does most of the work. On finishing, the state hurts: these are late steps where 73% of the options are FINISH and the LLM itself mostly wants to stop. Blind, the decider avoids finishing (below chance); with the state, it finishes while a required order is open more often than chance and more often than the LLM's own sample. The premature finishing is thus a misreading of the state, which agrees with, and strengthens, the LLM's own sense that the case is done.

### 4.4 Steering alone, the bias breaks the agent

Given the tool menu instead of proposals, the same model fails. S1-TOOL passes 7.0% of tasks and 44.3% of checkpoints ($\Delta$ $-34.0$ $[-45.0, -23.0]$ points in task success, $p < 0.001$). It retrieves

| Key error | Random | Blind | State | Blind |
|---|---|---|---|---|
| Order (97 q.) | 0.55 | 0.22 | 0.07 | +0.14 [0.08 |
| Redundancy (86) | 0.35 | 0.24 | 0.03 | +0.21 [0.11 |
| Finish (30) | 0.73 | 0.53 | 0.90 | −0.37 [−0.53, |

Table 6: **Blind test (P4), pplx-decider-v1.2.** Key errors of a random option, of the decider without the patient state (options only) and with it; differences with task-bootstrap 95% CIs. The LLM's own sample: 0.55, 0.28, 0.80.

| Tool | Clef (S1-TOOL) | LLM (LLM-TOOL) |
|---|---|---|
| Write note | 44 | 7 |
| Search labs | 17 | 31 |
| Search notes | 10 | 20 |
| Search problems | 4 | 9 |
| Place order | 7 | 4 |
| Search med. orders | 3 | 7 |
| Demographics | 5 | 5 |
| Finish | 4 | 4 |

Table 7: **Which tool each decider picks** under the same one-tool-per-turn restriction, as a share of steps (%), on the tasks both completed.

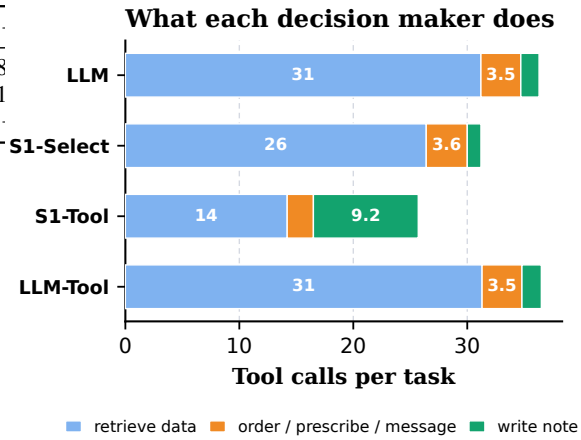



Figure 4: **Tool calls per task by kind.** Steering, the System One model under-retrieves and over-documents. LLM-TOOL: the LLM under the same one-tool-per-turn restriction, on the tasks it shares with S1-TOOL.

14.2 times per task against the LLM's 31.2, starts acting after 9.9 retrieval calls instead of 30.1, and writes the note 9.2 times, rewriting it again and again; it chooses to write the note at 44% of steps, where the LLM does so at 7% (Table 7). Restricting the LLM to one tool per turn costs nothing measurable (LLM-TOOL versus LLM: −0.5 [−4.7, +3.9] checkpoint points), while handing the same restricted choice to the System One model costs −36.5 [−43.2, −29.6] points. pplx-decider-v1.2 in the same seat falls behind as well (−24.7 [−34.7, −14.8] points over 49 tasks). In an open action space the commitment bias has nothing to anchor it: committing means writing, and the agent writes instead of looking.

### 4.5 Boundaries: where a System One decider is reliable

On the keyed exam questions we split the benefit of executing the decider's pick instead of the LLM's sample (key errors avoided per question) by what the decider and the LLM signal (Clef and pplx-decider-v1.2 shown; the other models behave alike).

**Its confidence is informative.** Overrides where the decider prefers its pick to the LLM's sample by a log-odds of at least 3 avoid 0.8–1.0 order-key errors per question and 0.6 redundancy errors; overrides below 1.5 avoid almost none. A margin keeps most of the benefit with far fewer overrides.

**It helps where the reasoning model is unsure.** When only 1–2 of the LLM's five samples are semantically the same decision as its first sample (same tools and key clinical arguments), the decider avoids 0.37–0.40 errors per question; at 4 of 5 it avoids 0.11–0.17, and at 5 of 5 none. Order decisions sit almost entirely at 1 of 5. The LLM's self-consistency and the decider's confidence are complementary signals for when to hand over.

**Finishing is its blind spot.** Where the decider confidently overrules the LLM's sample toward FINISH while a required order is still open, it is wrong in 4 of 5 cases (error change −0.75 to −0.80, $n = 4$–5), and every model finishes early more often than the LLM (Table 4). In S1-TOOL the same blindness appears as an agent that never stops writing.

**Long states.** 85% of keyed questions exceed the decider's 32k window and are read with shortened tool results. The deciders' error is 0.03–0.07 on the questions read in full and 0.20–0.23 on shortened ones, against 0.44–0.47 and 0.51–0.52 for the LLM's sample; long, truncated trajectories are where the System One view is weakest (confounded with late steps).

### 4.6 The same behaviour across six models

Is this Clef's behaviour, or how System One models behave as deciders? Table 8 gives every model's

| Decider | Order steps (answer key) | | | All questions | | |
|---|---|---|---|---|---|---|
| | Right | Orders, required | Orders, none req. | Agrees w/ LLM | Act | Note |
| Random | 45 | – | – | 22 | – | – |
| LLM (own sample) | 45 | 45 | 35 | 100 | – | – |
| LLM (most sampled) | 45 | – | – | – | – | – |
| Clef-27B | 94 | 98 | 66 | 23 | 15 | 10 |
| pplx-decider-v1.2 | 93 | 97 | 71 | 27 | 14 | 11 |
| d3 | 91 | 96 | 67 | 21 | 14 | 11 |
| linne-oak-1.0 | 92 | 95 | 53 | 37 | 13 | 11 |
| amos-1-large | 78 | 82 | 53 | 22 | 12 | 10 |
| decider-31b | 91 | 93 | 52 | 24 | 13 | 11 |

Table 8: **The same exam for every decider** (PhysicianBench, the LLM's own logged states and proposals; 1461 questions answered by every decider, 97 of them with an order key). Right: picks a required order on order steps where options differ. Orders, required / none req.: picks an ordering option when one places a required order, or when none does. Agrees w/ LLM: same pick as the LLM's own sample. Act, Note: share of all picks that place orders or write the note.

| Decider | S1-SELECT vs LLM | | S1-TOOL vs LLM-TOOL | |
|---|---|---|---|---|
| | Tasks | Δ checkpoints | Tasks | Δ checkpoints |
| Clef-27B | 100 | −1.1 [−4.7, +2.7] | 100 | −36.5 [−43.2, −29.6] |
| pplx-decider-v1.2 | 76 | +1.1 [−5.0, +7.2] | 49 | −24.7 [−34.7, −14.8] |

Table 9: **Whole tasks with another decider in Clef's seat.** Paired checkpoint differences with 95% intervals, each on the tasks both arms have completed.

answers to the fixed exam. On order steps all six pick a right option far more often than the LLM's own sample (Clef 94%, pplx-decider-v1.2 93%, d3 91%, linne-oak-1.0 92%, decider-31b 91%, amos-1-large 78%; the LLM 45%, random 45%), and all of them also order when no required order is on offer (53–71% of such steps, the LLM 35%). They agree with each other more than with the LLM (pplx-decider-v1.2 and d3 on –% of questions, Clef and pplx-decider-v1.2 on 50%). Online, pplx-decider-v1.2 in S1-SELECT keeps the LLM's checkpoint level (+1.1 [−5.0, +7.2] points over 76 tasks) and in S1-TOOL falls far behind (−24.7 [−34.7, −14.8] points over 49 tasks; Table 9). The commitment bias, its benefit on orders and its failure when steering are shared by the models we could test, not a quirk of Clef.

### 4.7 System One plus the reasoning LLM

The boundaries suggest a narrower role. Let the LLM gather data, write the note and decide when to stop; consult the System One model only where a proposal places an order, the decisions it gets right and the LLM defers. On the exam, this *orders-only* gate keeps the decider's order behaviour (pplx-decider-v1.2: omission 3%) and leaves finishing and redundancy at the LLM's level, while overruling the LLM on 11% of steps instead of 73%. Gating at every order or finish step overrules on 19%; a gate by log-odds margin or by the LLM's self-consistency does not lower the exam error further under cross-validation, because the exam's redundancy key rewards finishing over a note rewrite and the keys pull against each other. The exam cannot choose among these designs; whole runs must.

**Pre-registered online tests.** We fixed these tests before their data existed. P2′ is running at the time of writing, and we make no claim about the gate's online effect. **P1** replicated pplx-decider-v1.2 in S1-SELECT on the 24 tasks it had not run; prediction: more required-order checkpoints than the pooled LLM runs. It failed (−18 points [−36, −2], 16 tasks with such checkpoints; §4.1). **P2′** runs the orders-only gate with pplx-decider-v1.2; prediction: the required-order gain of S1-SELECT (which P1 has since failed to replicate) without its loss of notes (notes within the LLM's run-to-run noise, −0.12 [−0.40, 0.15]); documentation not lower; task pass reported with its CI, claimed only if the CI excludes 0. The blind test P4 is done (§4.3); it supports the design: the gate keeps the decider where its preference is a useful prior (orders) and takes it away where reading the state misleads it (finishing).

## 5 Related work

**Choosing among samples.** Majority voting over reasoning paths (Wang et al., 2023), trained verifiers (Cobbe et al., 2021), process reward models (Lightman et al., 2024) and LLM judges (Zheng et al., 2023) all pick among sampled outputs. S1-SELECT applies best-of-$K$ selection at every step of an agent loop, with a decision model as the selector; our gated variant uses the LLM's own self-consistency to decide when to consult it.

**System One decision models.** JEV introduced the typed-question interface (Almeida, 2026); Clef is an open 27B model with a joint head over a Qwen3.8-27B backbone (Cloudflare, 2026). As judges they approach frontier LLMs on verdicts that can be read off the input at a small fraction of the cost (Li et al., 2026b); as next-action selectors in general agents, most of their wrong actions follow a correct reading of the state (Li et al., 2026a). We bring six of them into a clinical agent and characterise a shared bias.

**Clinical agent benchmarks.** PhysicianBench grades long-horizon physician tasks in a live record system by code and rubric (Liu et al., 2026). It has been run only with generative LLM agents.

## 6 Discussion

**Deciding is not the bottleneck.** Replacing the component that picks the next turn is the natural place to put a fast decision model, and it is safe there, but it is not where a capable LLM agent loses its points. The LLM mostly defers rather than omits, and what it never proposes no selector can choose. A decider that commits earlier trades the few orders it recovers for orders the LLM would have proposed after more retrieval. Gains in clinical agents of this kind must come from generation and information gathering, or from a decider that also controls when the LLM keeps looking.

**A commitment bias: a prior and a misreading.** The clearest regularity in our data is a preference over kinds of action that six independently trained System One models share. It explains when they help (the LLM defers orders it already proposes) and when they hurt (finishing early, writing instead of looking). The blind test splits it: the preference for orders is largely a prior over the options, while the excess of premature finishing comes from reading a state in which the work looks done, the same reading the LLM makes, held more firmly.

**A division of labour.** The LLM is good at what the System One model cannot do: deciding what to look up, how much evidence is enough, when to stop, and writing the note. The System One model is good at what the LLM does unreliably: committing to a concrete order once it is on the table. The design that follows consults the System One model at order steps only, and preferably where the LLM's samples disagree and the System One model is confident. The ceiling above bounds what it can gain; its value may lie less in outcome than in cost and in a stable, inspectable decision at the steps that commit.

**Evaluating deciders inside agents.** Step-level keys built on an agent's own trajectory reward timing and style. Their offline gains overstated the online effect by an order of magnitude here, and even an online gain on 76 tasks did not survive 24 more. Deciders should be ranked on whole runs, with several runs per arm and a held-out replication, because the run-to-run noise of a single LLM policy spans $\pm 10$ points of task success. A blind control (the same questions without the state) separates what a decider reads from what it prefers.

**Cost.** The System One decision is one forward pass, about 4.7 s per step on one GPU, and adds no generated tokens. The cost of S1-SELECT lies in the five proposals (119k versus 29k generated tokens per task); the orders-only gate needs proposals only where an order is on the table.

## Limitations

We study one LLM and one benchmark. Clef shares its backbone with the LLM; the five other System One models are evaluated on the exam and, for one of them, online. Most online numbers come from one run per arm (the LLM has 2.15 runs per task when pooled), and the online test of the gated design is incomplete. The benchmark barely penalises over-ordering, so a commitment bias costs little here and could cost more in practice. Our judge is the local LLM rather than the leaderboard's, so absolute scores are not comparable with the leaderboard, while every arm shares the judge. The System One models read a 32k-token view of states the LLM reads in full. Inpatient settings such as ClinEnv (Lu et al., 2026) are left for future work.

## Ethics and data

PhysicianBench's records are real de-identified patients under a data use agreement. All models run locally, a socket guard blocks any non-local connection from agent and grader processes, and trajectories and notes stay on access-controlled storage outside version control. We report aggregates only.

## References


D. Almeida. 2026. Introducing system one models & jev. TypeSafe AI blog, https://typesafe.ai/blog/introducing-system-one-models-and-jev.

Cloudflare. 2026. Clef decision models. Blog post and model card, https://blog.cloudflare.com/clef-decision-models. Model: Cloudflare/clef (27B, post-trained from Qwen3.8-27B).

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *Preprint*, arXiv:2110.14168.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*.

Yubo Li, Ramayya Krishnan, and Rema Padman. 2026a. Reading right, acting wrong: Diagnosing decision failures of system one models in agent loops. *Preprint*, arXiv:2610.18579.

Yubo Li, Yidi Miao, Ramayya Krishnan, et al. 2026b. JEV-as-a-judge: Accept when confident, escalate when unsure. *Preprint*, arXiv:2609.26550.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let's verify step by step. In *International Conference on Learning Representations*.

Ruoqi Liu, Imran Q. Mohiuddin, Austin J. Schoeffler, Kavita Renduchintala, Ashwin Nayak, Prasantha L. Vemu, Shivam C. Vedak, Kameron C. Black, John L. Havlik, Isaac Ogunmola, Stephen P. Ma, Roopa Dhatt, and Jonathan H. Chen. 2026. Physicianbench: Evaluating llm agents on physician tasks in real-world ehr environments.

Lu, Lin, Shi, Tamo, Zhao, Wang, and Wang. 2026. ClinEnv. *Preprint*, arXiv:2606.02568. TODO: verify full title and author first names.

Qwen Team. 2026. Qwen3.8-27B. Model card, `https://huggingface.co/Qwen/Qwen3.8-27B`.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*.

## A Decision interface

**Questions.** The System One model receives the state and one Choice question. For S1-SELECT: *"The state shows a clinical task assigned to an AI agent that works in an electronic health record through FHIR tools, followed by the agent's actions and their results so far. Each option is a candidate next action for the agent. Choose the candidate that best advances completing the task correctly, completely and safely."* For S1-TOOL the second and third sentences read: *"Each option is a tool the agent can use for its next action, or finishing with a final answer. Choose what the agent should do next to complete the task correctly, completely and safely."* Options are the rendered turns (S1-SELECT) or each tool's name with the first paragraph of its description, plus FINISH (S1-TOOL).

**State.** The state is the task instruction followed by every step: the action, rendered as tool calls with full arguments, and its results as the LLM saw them. It is fitted into the System One model's 32,768-token context deterministically: tool results are kept in full, then shortened oldest-first to 800 characters, then omitted oldest-first, each by the smallest number that fits. In the S1-SELECT run 44% of decisions saw shortened results and 1% saw omitted ones.

**Other System One models.** pplx-decider-v1.2 (`cce0f3f8`), d3 v3.1.0 (`cbff3a3c`), linne-oak-1.0 (`20da7bf3`), amos-1-large (`d67e051c`) and decider-31b (`e12ddf20`) each run their authors' own serving code on H100 GPUs, locally and offline. pplx-decider-v1.2 ships a library only, which we call on its reference path. decider-31b is NVFP4, which an H100 runs through vLLM's weight-only FP4 path; its authors used Blackwell GPUs. linne-oak-1.0 needs two GPUs for 32k-token inputs. Every model receives the state and question built for Clef, with the same fitting rule, counted in its own tokenizer. Five exam questions whose options alone exceed the 32k window are unanswerable for every model.

**The LLM.** Qwen3.8-27B with thinking on, temperature 1.0, top-$p$ 0.95, top-$k$ 20, at most 16,384 completion tokens per call. Identical replicas serve all arms; each task is routed whole to the less loaded replica, independent of the arm.

## B Additional results

Table 10 lists every paired comparison with intervals; Table 11 describes the decisions. Episodes that ended without a final answer (step limit, repeated-call abort or context overflow) number 2 for LLM, 1 for S1-SELECT and 16 for S1-TOOL

| Comparison | Tasks | Pass@1 | Δ pass@1 [95% CI] | $p$ | Δ checkpoints [95% CI] | Δ DR | Δ CR | Δ AE | Δ Doc |
|---|---|---|---|---|---|---|---|---|---|
| S1-SELECT vs LLM | 100 | 37.0 / 41.0 | −4.0 [−13.0, +5.0] | 0.52 | −1.1 [−4.7, +2.7] | +1.8 | −1.2 | +5.1 | −5.1 |
| S1-TOOL vs LLM-TOOL | 100 | 7.0 / 35.0 | −28.0 [−38.0, −18.0] | <0.001 | −36.5 [−43.2, −29.6] | −39.3 | −36.9 | −27.4 | −38.4 |
| LLM-TOOL vs LLM | 100 | 35.0 / 41.0 | −6.0 [−17.0, +5.0] | 0.38 | −0.5 [−4.7, +3.9] | +0.8 | +0.4 | +4.7 | −6.1 |

Table 10: Paired comparisons, each on the tasks both arms completed (run 1). Δ in points; per-type columns give the point difference in checkpoint success.

| | Steps /task | Distinct strings (%) | Distinct actions (%) | Mean max $p$ | Norm. entropy | Departs from LLM sample (%) | Decide (s) |
|---|---|---|---|---|---|---|---|
| S1-SELECT | 11.8 | 99 | 89 | 0.382 | 0.886 | 79 | 4.7 |

Table 11: S1-SELECT decisions, run 1. Distinct strings: steps whose five proposals are not all identical. Distinct actions: after merging proposals with the same tools and key clinical arguments. Mean max $p$ and entropy (normalised by $\log n$) describe the System One model's distribution where there was a choice. Departs: share of those steps where it chose a proposal other than the LLM's first sample.

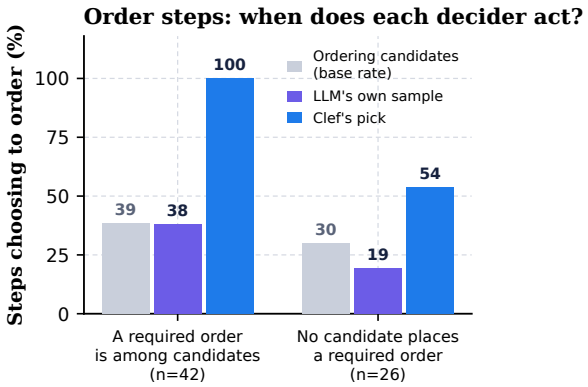



Figure 5: **Clef places required orders when they are on the table.** Order steps of the S1-SELECT run whose proposals mix ordering and other turns. Orders the graders do not require are not wrong, only uncredited; most of the LLM's skipped orders are placed at a later step (§4.2).

on the paired tasks. Distinct proposals are semantically different decisions at 89% of S1-SELECT steps; on the exam, 952 of 1,460 questions are choices among retrieval turns only, which no key scores.